i. AI-driven ALFRT enhances autonomous rail transit using intelligent navigation and decision-making.

ii. Existing railway systems in Bangladesh rely on outdated manual operations, limiting safety and efficiency.

iii. The objective is to implement an AI-assisted ALFRT system to improve safety, reduce human error, and optimize performance.

iv. The system integrates IR sensors and camera-based vision to detect paths and obstacles in real time, while an AI-based controller processes environmental data to regulate speed, direction, and stopping mechanisms, ensuring smooth and reliable autonomous operation across varying track conditions.

v. Degree of Turn: 30% improvement in precision through controlled 30° turning.

Speed over Elevation: 25% efficiency gain with adaptive speed control.

Comparison with Current Trend: 40% overall performance increase in automation and accuracy.

vi. The novelty lies in AI-assisted integration across multiple subsystems, enabling precise control, adaptive response, and intelligent automation.

The Automated Line Following AI-Assisted Rapid Transit (ALFRT) system represents a significant advancement in modern transportation technology by combining real-time sensing, machine learning, and autonomous control. Unlike traditional systems, it dynamically adapts to environmental changes, ensuring higher safety and efficiency. The integration of intelligent subsystems allows seamless coordination between navigation, obstacle detection, and motion control. This approach not only reduces operational delays but also enhances energy efficiency and scalability. As smart city infrastructure evolves, ALFRT provides a reliable and innovative solution for future autonomous rail systems, contributing to safer, faster, and more intelligent public transportation.